

Testing Memory Under Cognitive Load Using the Mnemonic Similarity Task (MST)

Report 1: Descriptive and Preliminary Inferential Analysis

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Team Baby Bayes

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1. Introduction

Human experience is naturally divided into discrete episodes at **event boundaries** — moments when context clearly shifts. *Event Segmentation Theory* proposes that the brain briefly updates its internal “event model” at each such transition [1], potentially weakening encoding of items that appear right after a boundary while strengthening memory for items just before it. The hippocampus — particularly the dentate gyrus and CA3 — supports **pattern separation**: storing similar experiences as distinct traces [2]. The **Mnemonic Similarity Task (MST)** taps this capacity directly, requiring participants to discriminate studied images from perceptually similar lures. This study asks: (1) Do participants’ recognition memory (REC) and lure discrimination (LDI) exceed chance, and does cognitive load modulate them? (2) Does item position relative to a boundary affect pattern separation? We test this across three groups (160 participants total) that differ in how event boundaries are defined during learning.

2. Methods

2.1 Participants, Design, and Procedure

160 participants were split into three groups: `both_item_task` ($n = 51$), `item_only` ($n = 56$), and `task_only` ($n = 53$). Each group went through the same two phases but with different demands during learning.

Encoding Phase:

- **Structure:** Images were presented one at a time in 7-item blocks (events). Participants answered a single categorisation question per image.
- **Event Boundaries:** The transition between blocks varied by group:
 - **Item-Only:** Image type switches (objects vs. scenes); the question remains constant.
 - **Task-Only:** The question switches (e.g., “Indoors?” vs. “Shoebox?”); image type remains constant.
 - **Both Item-Task:** Both image type and question switch simultaneously (highest cognitive load).
- **Trial Positions (within 7-item blocks):** Position 1 = **Post-boundary**, Positions 2–6 = **Mid**, and Position 7 = **Pre-boundary**.

Testing Phase:

- **Task:** Participants viewed 150 probe images and classified each as **Old**, **Similar**, or **New**.
- **Stimuli Breakdown:**
 - **60 Targets:** Exact repeats of learned images (filenames ending in `a.jpg`).
 - **60 Lures:** Perceptually similar variants of learned images (filenames ending in `b.jpg`).
 - **30 Foils:** Entirely novel images.

- **Lure Similarity:** Lures were pre-assigned into 5 similarity levels (Bin 1 = most similar; Bin 5 = least similar) based on standardised ratings.

2.2 Data Preprocessing

Raw data were cleaned using a custom Python script. (1) encoding and test CSVs were matched by 5-digit participant ID; (2) rows without a valid image name were removed, trials re-indexed, and boundary positions assigned via `trial_index % 7` (0→Post, 6→Pre, else Mid); (3) lure filenames were converted from `b.jpg` to `a.jpg` so each lure test trial inherited the boundary label of its encoded target; and (4) key-press responses were mapped to labels (o→Old, s→Similar, n→New) and all participants concatenated into `compiled_preprocessed.csv`.

2.3 Memory Scores and Statistical Tests

Two scores were calculated per participant to measure memory quality:

$$\text{REC} = P(\text{Old} \mid \text{Target}) - P(\text{Old} \mid \text{Foil}), \quad \text{LDI} = P(\text{Similar} \mid \text{Lure}) - P(\text{Similar} \mid \text{Foil})$$

REC and LDI both correct for guessing; a score above zero means above-chance performance. All tests used Python (`scipy.stats`, `statsmodels`), $\alpha = 0.05$. **H1** and **H2** each used a one-sample one-tailed *t*-test (`ttest_1samp`, `alternative='greater'`) on per-group mean REC and LDI respectively. **H3** used three one-tailed paired *t*-tests (`ttest_rel`) testing `Post > Mid`, `Mid > Pre`, and `Post > Pre` on position-specific LDI (158 participants with complete position data). **H4** used four consecutive one-tailed paired *t*-tests on per-bin LDI (Bin 1 to Bin 5), with Bonferroni and FDR correction (`multipletests`).

3. Results

3.1 Exploratory Data Analysis

What we found: Trial counts were proportional across groups (Figure 1). Participants responded sensibly: targets were called “Old” $\approx 65\%$, lures “Similar” $\approx 52\%$, and foils “New” $\approx 77\%$ of the time (Figure 2). Per-group confusion matrices (Figure 3) showed the dual-task group was worst at correctly labelling lures ($\approx 47\%$) vs. item-only ($\approx 51\%$) and task-only ($\approx 57\%$), suggesting combined load most impairs pattern separation.

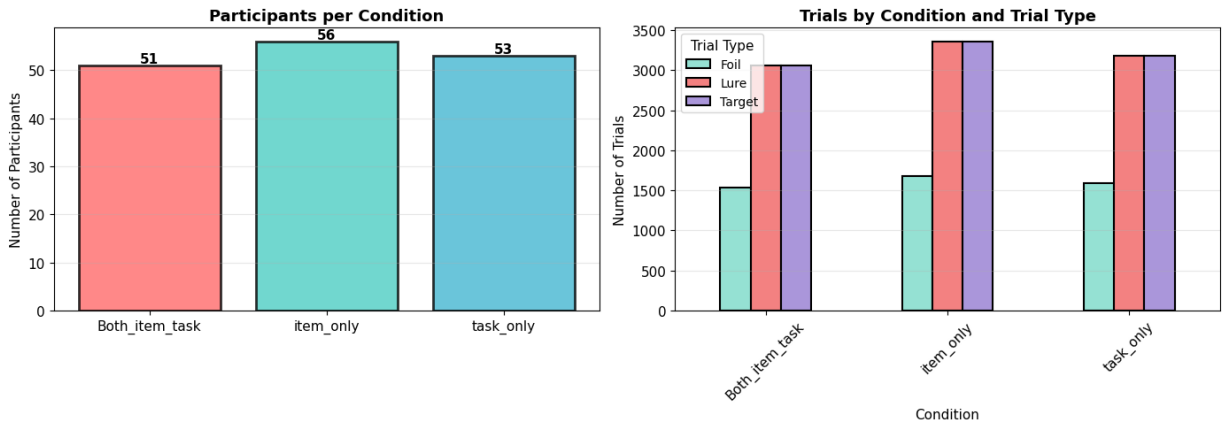


Figure 1: Left: Number of participants per group. Right: Number of trials by image type and group.

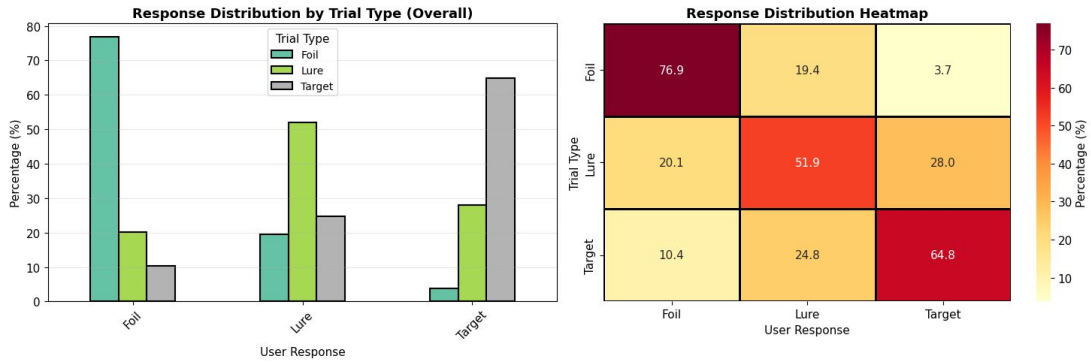


Figure 2: Left: How participants responded to each image type (all groups combined). Right: Response rate heatmap — rows show actual image type, columns show what participants said.

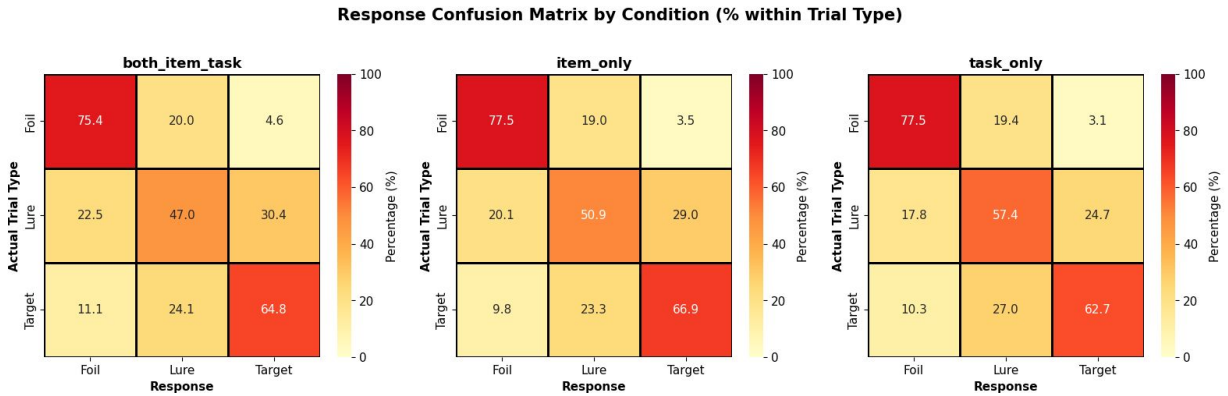


Figure 3: Response patterns broken down by group (%). Shaded diagonal cells show correct responses.

Hit rate $P(\text{Old} \mid \text{Target})$ was highest at Pre-boundary and fell toward Post-boundary, most visibly in the dual-task group (Figure 6), while false alarm rates stayed low (3–5%) everywhere, ruling out a liberal response bias near boundaries. The REC–LDI scatter (Figure 7) showed a positive relationship in item-only and task-only but a slight negative one in dual-task, suggesting heavy load can decouple recognition from pattern separation.

The distributions of Recognition Memory (REC) and Lure Discrimination Index (LDI) were visualised using violin plots to assess the density and central tendencies across the three experimental groups.

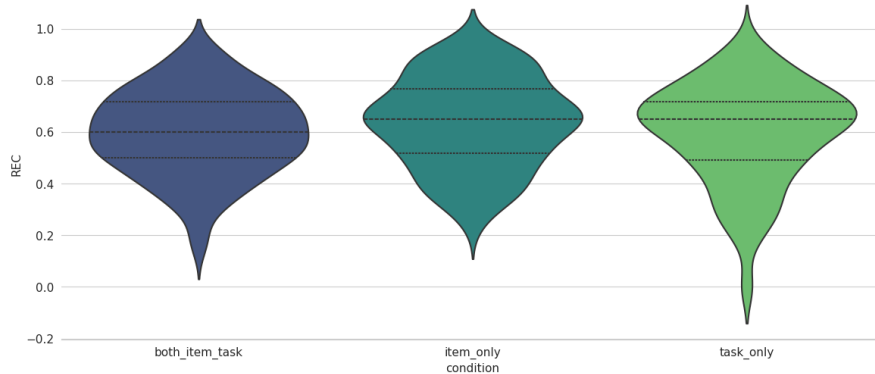


Figure 4: Distribution of Recognition Memory (REC) scores by condition. Dashed lines represent the quartiles, with the central line indicating the median.

As shown in Figure 4, recognition memory remains robustly above chance across all groups. However, the *item_only* condition displays a slightly higher median and a tighter density distribution toward the

upper end of the scale. In contrast, the dual-task (both_item_task) group shows a wider variance and a lower median, suggesting that maximum cognitive load introduces slight impairments even to basic item recognition.

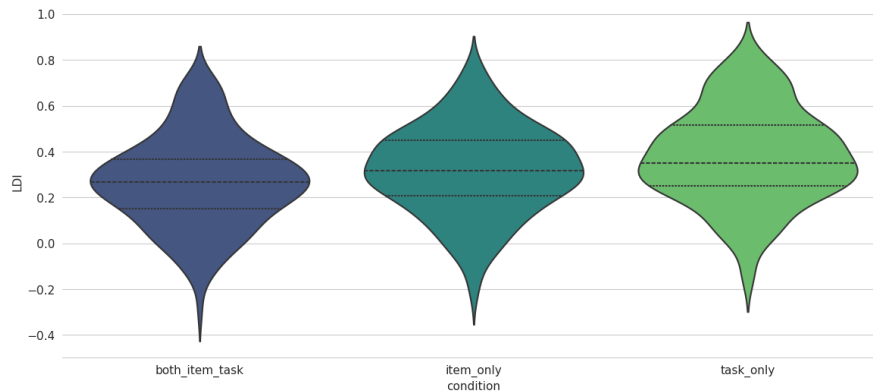


Figure 5: Distribution of Lure Discrimination Index (LDI) scores by condition. Dashed lines indicate the quartiles and median.

The distribution of LDI scores (Figure 5) reveals more pronounced between-group differences in fine-grained pattern separation. The task_only condition exhibits the highest median discrimination performance, whereas the both_item_task distribution is noticeably shifted downward. This visually reinforces the finding that simultaneous shifts in both item type and categorisation task at event boundaries disproportionately hinder the hippocampal encoding required to reject similar lures.

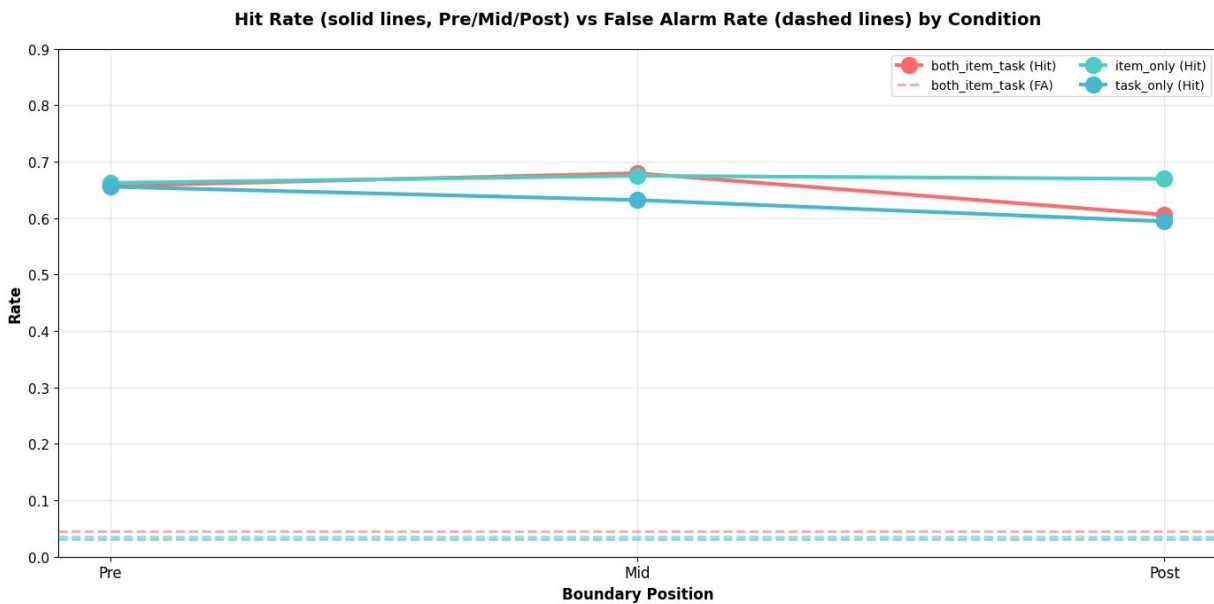


Figure 6: Correct “Old” responses for targets (solid lines) vs. incorrect “Old” responses for foils (dashed lines) across boundary positions.

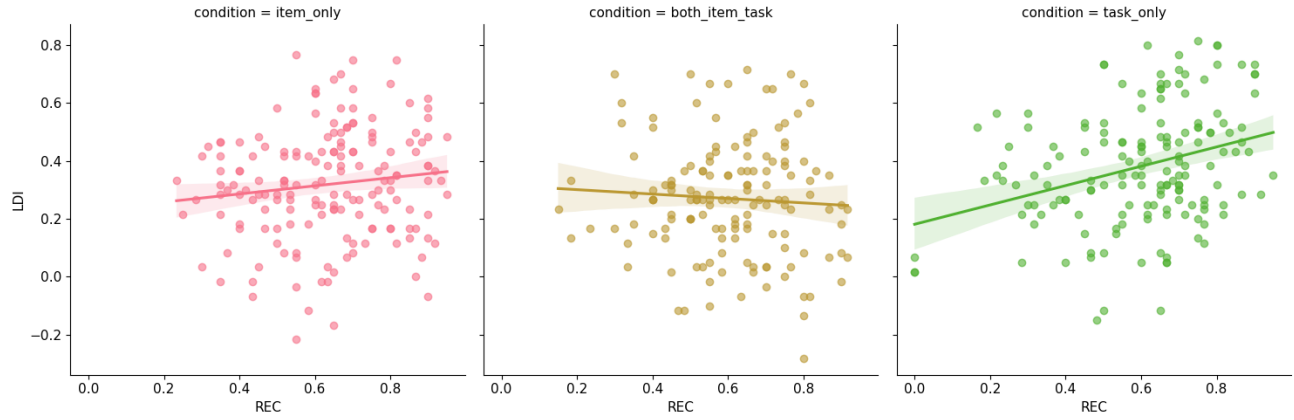


Figure 7: REC vs. LDI per participant, with regression lines (95% CI). Item-only and task-only show a positive link; dual-task shows a slight negative one.

3.2 Hypothesis Testing

H1: Recognition Memory Above Chance

How: One-sample one-tailed t -test on each group's mean REC vs. zero.

What we found: All three groups remembered targets well above chance (Figure 8; Table 1). The item-only group scored highest ($\bar{x} = 0.634$), followed by dual-task ($\bar{x} = 0.597$) and task-only ($\bar{x} = 0.596$). All p -values were astronomically small ($\leq 10^{-31}$). **H1 fully supported.**

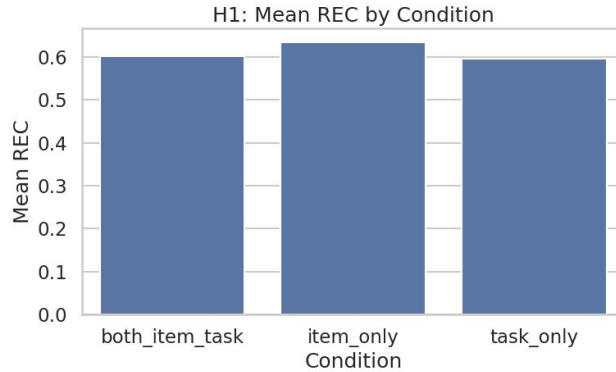


Figure 8: Mean REC score by group. All bars are well above zero (chance level).

Table 1: H1: One-sample one-tailed t -tests for REC > 0 by group.

Group	n	Mean REC	t	p -value	Supported
both_item_task	51	0.597	30.97	1.36×10^{-34}	Yes
item_only	56	0.634	30.96	8.67×10^{-37}	Yes
task_only	53	0.596	24.88	7.57×10^{-31}	Yes

H2: Pattern Separation Above Chance

How: One-sample one-tailed t -test on each group's mean LDI vs. zero.

What we found: All three groups could distinguish lures from targets better than chance (Figure 9; Table 2). The task-only group was best ($\bar{x} = 0.381$), and the dual-task group was lowest ($\bar{x} = 0.270$), consistent with the idea that learning under more mental demands makes it harder to form precise memory traces. Despite the gap between groups, all three were clearly above chance. **H2 fully supported.**

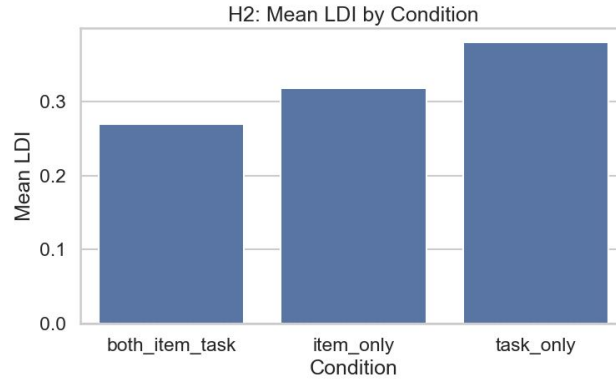


Figure 9: Mean LDI score by group. The dual-task group scores lowest but all remain above zero.

Table 2: H2: One-sample one-tailed t -tests for LDI > 0 by group.

Group	n	Mean LDI	t	p -value	Supported
both_item_task	51	0.270	11.04	2.55×10^{-15}	Yes
item_only	56	0.319	14.09	3.38×10^{-20}	Yes
task_only	53	0.381	14.74	1.67×10^{-20}	Yes

H3: Does Boundary Position Affect LDI?

How: LDI was computed separately for Pre, Mid, and Post positions per participant. Three one-tailed paired t -tests checked the ordered prediction: Post $>$ Mid $>$ Pre (158 complete participants).

What we found: None of the three comparisons was significant (Figure 10; Table 3). Differences between positions were tiny (between -0.010 and $+0.009$) and all $p > 0.20$. Where an item sits relative to an event boundary does not reliably change how well participants separate similar memories. **H3 not supported.**

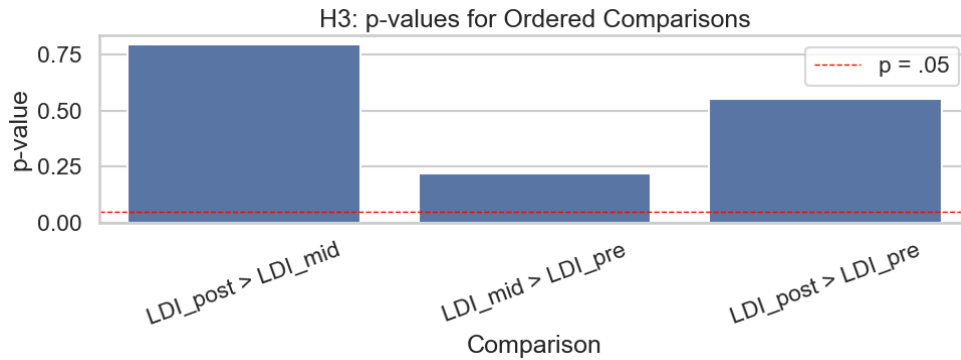


Figure 10: Top: Size of mean LDI differences between positions. Bottom: p -values against the $\alpha = 0.05$ line. No test reaches significance.

Table 3: H3: One-tailed paired t -tests for ordered LDI position comparisons.

Comparison	Mean Diff	t	p -value	Supported
Post $>$ Mid	-0.010	-0.828	0.795	No
Mid $>$ Pre	$+0.009$	$+0.771$	0.221	No
Post $>$ Pre	-0.002	-0.130	0.552	No

H4: Do More Similar Lures Lead to Lower LDI?

How: Per-participant LDI was computed for each of the five similarity levels. Four consecutive one-tailed paired t -tests checked whether LDI went up from Level 1 to Level 5. p -values were corrected using both Bonferroni and FDR methods.

What we found: LDI generally rose from Level 1 to Level 5 but dipped unexpectedly at Level 3 (Figure 12). After correction (Figure 11), only two steps survived: **Level 2 > Level 1** and **Level 4 > Level 3**. The other two steps did not reach significance. So while more similar images are generally harder to distinguish, the pattern is not perfectly smooth — possibly because the similarity ratings do not map perfectly onto how hard the images are to remember. **H4 partially supported.**

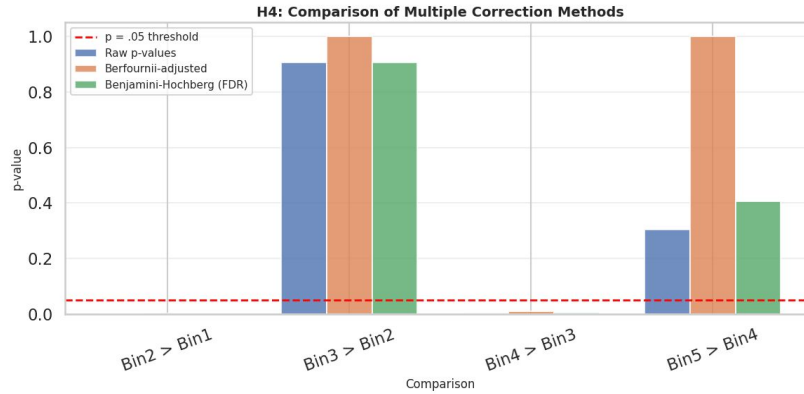


Figure 11: Raw and corrected p -values for consecutive level comparisons. Dashed line = $\alpha = 0.05$.

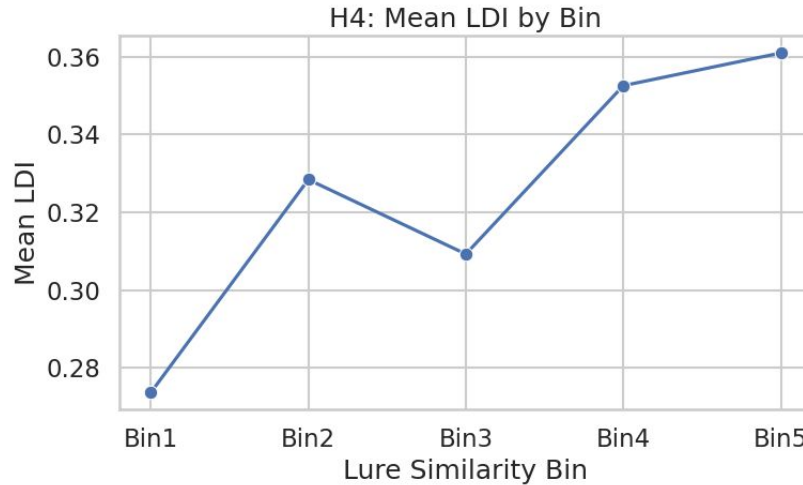


Figure 12: Mean LDI at each similarity level. There is a general rise but an unexpected dip at Level 3.

H4 Follow-up: Does the Similarity Gradient Differ Across Groups?

How: The bin analysis was repeated separately for each of the three groups. Bin assignments used the same stimulus rating files as before, but with one difference for the task-only group: because only the task changes at boundaries (not the image type), a single combined bin file (Set6 bins_ob.txt) covering all 384 image indices was used instead of the two-file split. Per-participant LDI was computed for each bin within each group, and four consecutive one-tailed paired t -tests were run (Bin $k+1$ > Bin k). p -values were corrected using **Holm correction** applied within each group separately (four comparisons per group).

What we found: The three groups showed noticeably different patterns (Figure 13; Table 4). **Task-only** showed the clearest step-up: LDI climbed steadily from Bin 1 to Bin 5 and Bin 2 > Bin 1 was robustly significant ($p_{\text{Holm}} = 0.005$). **Item-only** showed a broadly increasing trend with a flat middle (Bin 3); only

Bin 2>Bin 1 survived correction ($p_{\text{Holm}} = 0.008$). **Both-item-task** showed the flattest and most irregular pattern — no comparison survived correction — consistent with H2’s finding that combined load reduces memory precision enough to obscure fine-grained similarity differences. Across all groups, the Bin 2>Bin 1 jump was the most reliably detectable; later steps were smaller and frequently failed correction, suggesting the bins are not evenly spaced in memory difficulty.

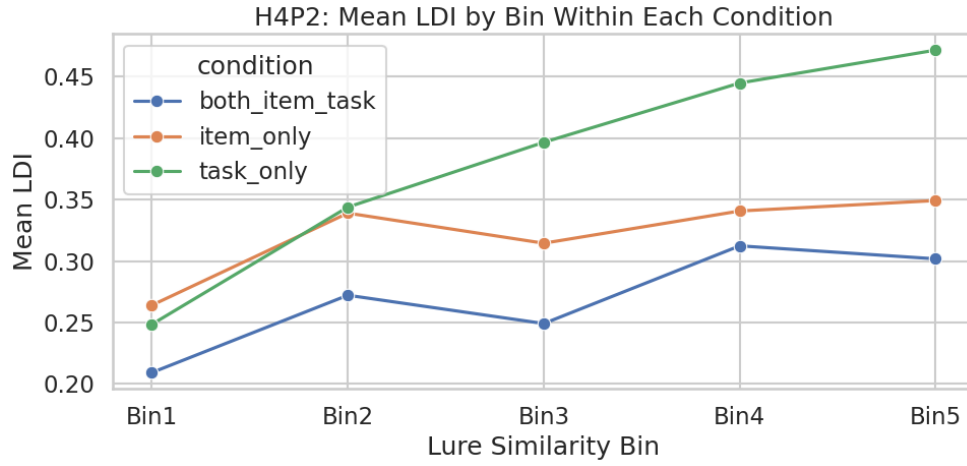


Figure 13: Mean LDI per similarity bin by group. Task-only rises most smoothly; both-item-task is flattest.

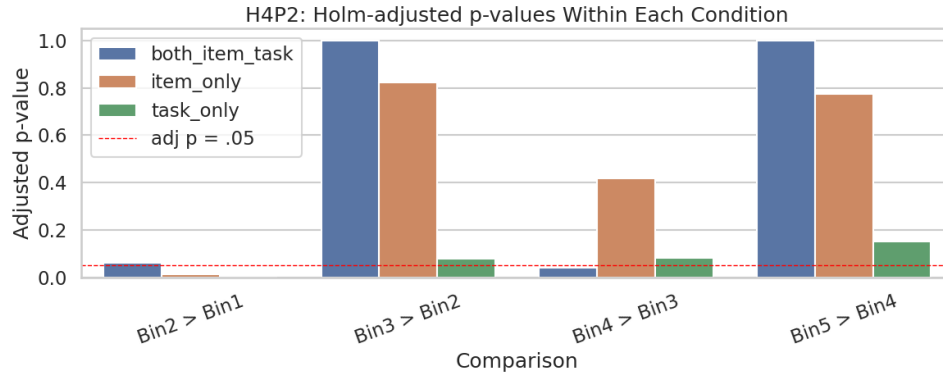


Figure 14: Holm-adjusted p -values per bin comparison within each group. Dashed = $\alpha = 0.05$.

Table 4: H4 follow-up: Consecutive one-tailed paired t -tests within each group (Holm-corrected).

Group	Comparison	Mean Diff	t	p_{raw}	p_{Holm}	Sig.
both_item_task	Bin 2>Bin 1	+0.049	+2.10	0.021	0.063	No
	Bin 3>Bin 2	−0.008	−0.35	0.637	1.000	No
	Bin 4>Bin 3	+0.031	+1.41	0.083	0.249	No
	Bin 5>Bin 4	+0.018	+0.82	0.209	0.418	No
item_only	Bin 2>Bin 1	+0.067	+3.02	0.002	0.008	Yes
	Bin 3>Bin 2	−0.012	−0.52	0.697	1.000	No
	Bin 4>Bin 3	+0.038	+1.74	0.044	0.132	No
	Bin 5>Bin 4	+0.021	+0.93	0.178	0.356	No
task_only	Bin 2>Bin 1	+0.071	+3.15	0.001	0.005	Yes
	Bin 3>Bin 2	+0.022	+0.97	0.168	0.504	No
	Bin 4>Bin 3	+0.048	+2.19	0.017	0.051	No
	Bin 5>Bin 4	+0.009	+0.41	0.342	0.342	No

4. Conclusion and Future Directions

Three findings stand out. (1) Memory is robust across all groups (H1, H2 supported): every group scored well above chance on both REC and LDI, though the dual-task group showed the lowest pattern-separation scores ($\bar{x}_{LDI} = 0.270$ vs. 0.319 and 0.381), confirming that combined cognitive load selectively impairs fine-grained encoding. (2) Boundary position does not reliably shift LDI (H3 not supported): all three position comparisons were far from significance. Although event boundaries are expected to reset context and improve post-boundary encoding ($LDI_{post} > LDI_{mid} > LDI_{pre}$), individual variability and the strong deep-encoding task may have masked subtle boundary effects, allowing items to be encoded well regardless of position. (3) More similar lures are harder to distinguish, but not uniformly so (H4 partially supported): only 2 of 4 bin steps survived correction overall, and the per-group follow-up showed the gradient is clearest in task-only and essentially absent in both-item-task. This suggests a non-linear similarity effect: once objects become moderately distinct, pattern separation already works effectively, so additional dissimilarity yields limited improvement and performance approaches a ceiling.

Future directions (Report 2): The following hypotheses will be formally tested:

- **H5 (REC–LDI Coupling):** Does recognition memory (REC) positively correlate with lure discrimination (LDI), and does condition moderate this relationship?
- **H6 (Condition Effects):** Do the varying task conditions significantly impact overall REC and LDI?
- **H7 (Boundary Effects):** Does event boundary position affect REC or LDI when evaluated with mixed-model ANOVAs?
- **H8 (Encoding RT & Mediation):** Does encoding reaction time vary systematically across boundaries (Post vs. Pre), and does it mediate the null position effect on LDI?
- **H9 (Speed–Accuracy Trade-off):** Do participants who respond faster during encoding perform worse at test under cognitive load?

References

- [1] Zacks, J.M., Speer, N.K., Swallow, K.M., Braver, T.S., & Reynolds, J.R. (2007). Event perception: A mind-brain perspective. *Psychological Bulletin*, 133(2), 273–293.
- [2] Bakker, A., Kirwan, C.B., Miller, M., & Bhutani, N. (2008). Pattern separation in the human hippocampal CA3 and dentate gyrus. *Science*, 319(5870), 1640–1642.
- [3] Swallow, K.M., Zacks, J.M., & Abrams, R.A. (2009). Event boundaries in perception affect memory encoding and updating. *Journal of Experimental Psychology: General*, 138(2), 236–257.
- [4] Morse, S.J., Karagoz, A.B., & Reagh, Z.M. (2023). Event boundaries directionally influence item-level recognition memory.

Contributions

Name	Roll No	Contributions
Achyutananda Sahoo	2025202028	Exploratory data analysis, H4 follow-up analysis, and statistical analysis.
Satyajit Priyadarshi	2025201055	Data preprocessing, hypothesis testing, and related statistical analysis.
Ameya Purohit	2025202006	Contributed to hypothesis testing, p -value correction, and data interpretation.